



Development Case Study

Shariah-Compliant AI Robo-Advisor Pipeline

September 25, 2026

Roadmap

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- 3 The AI Forecasting Attempts
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The Problem

Objective

Build a Shariah-compliant robo-advisor for Bursa Malaysia that:

- Screens a universe of listed equities against Shariah criteria
- Estimates return and risk per asset
- Constructs a mean-CVaR optimal portfolio subject to transaction costs, holding limits, and zakat

The Question We Actually Wanted to Answer

Can modern AI — neural networks, deep sequence models, gradient boosting — generate a return forecast with demonstrable out-of-sample skill on a small, heterogeneous, emerging-market equity universe?

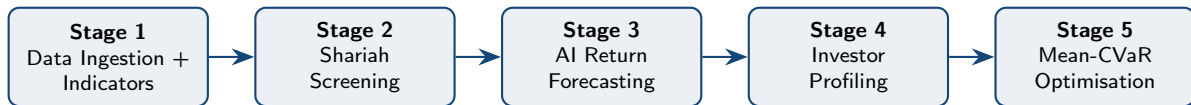
On this universe, at this horizon, with publicly available features:

No AI model beat the naive baseline out-of-sample.

Three architectures were tested across three rounds of engineering. All three collapsed to predicting the historical mean.

This is a finding, not a failure.

The Original Six-Stage Pipeline



- Universe: 25 Bursa Malaysia large-caps across 15 sectors
- 5-year window: 1,227 trading days $\times$ 25 tickers = 30,675 panel rows
- 21 technical indicators engineered per ticker
- **Stage 3 is where the trouble lived.** Everything else worked.

Stage 1 — Data Ingestion

What we pull

- Daily OHLCV via `yfinance`
- Balance-sheet items: total assets, total debt, cash, receivables
- Industry classification (for Shariah screening)

Live data sample

- 25/25 tickers returned
- 1,227 trading days each
- 30,675 panel rows

Fallback path

- If live pull fails, generate a schema-identical synthetic dataset (GBM + noise)
- This made the pipeline *look* robust while masking bugs
- **Lesson: synthetic fallbacks hide live-data pathologies**

Stage 2 — Shariah Screening

Tier 1: Business Activity

Exclude on sector / industry:

- Conventional banking and insurance (*riba*)
- Gambling and casinos (*maysir*)
- Breweries, distilleries, tobacco
- Weapons, adult entertainment, non-halal food

Tier 2: Financial Ratios (SC Malaysia-style)

$$\frac{\text{cash \& interest-bearing securities}}{\text{denominator}} < 0.33 \quad \frac{\text{interest-bearing debt}}{\text{denominator}} < 0.33 \quad \frac{\text{receivables + cash}}{\text{denominator}} < 0.50$$

Stage 2 — Screening Result

Tier 1 exclusions (6/25):

- 1155.KL Maybank
- 1295.KL Public Bank
- 3182.KL Genting
- 4715.KL Genting Malaysia
- 2836.KL Carlsberg
- 3255.KL Heineken

13 / 25 pass both tiers.

A Warning Sign (retrospective)

These six exclusions only fired correctly *after* a major bug fix. In the first run, Maybank and Carlsberg silently passed. We will return to this.

Three architectures. Three feature sets. One question:

Does anything beat “predict the historical mean”?

Round	Model	Features	Verdict
1	Per-asset MLP	19 technicals	0/17
2	CNN-LSTM	19 tech. + 6 fundamental	0/13
3	Cross-sectional GBM	19 tech. + 6 fundamental	0/13

Architecture

- `MLPRegressor(hidden=(32,16))`
- `StandardScaler` on features
- Per-ticker fit
- Target: 21-day forward log return
- 80/20 train/test split

Hypothesis

- Technical indicators contain predictive signal for short-horizon returns

Result

- **10 of 17** tickers produced forecasts pinned at the $\pm 60\%$ annualised clip
- Raw ANN outputs reached $\pm 500\%$
- `val_rmse` exceeded target std on most tickers
- The clip was the only reason the column looked bounded

Round 1 — Saturation Symptom

Ticker	exp_return_ann	val RMSE
2836.KL	0.6000	0.1212
5168.KL	0.6000	0.4508
3255.KL	0.6000	0.1368
8869.KL	0.6000	0.3943
5211.KL	0.6000	0.1000
3689.KL	0.1806	0.1808
4707.KL	0.1089	0.1192
...	...	...
1155.KL	-0.6000	0.2099
3026.KL	-0.6000	0.1101

Diagnosis

10/17 forecasts are exactly ± 0.60 — the guard-rail value. The model has no predictive power; the clip is the only thing making the output look reasonable.

Round 2 — CNN-LSTM + Fundamentals

Architecture

- 1-D CNN over feature axis (local interactions)
- LSTM over time axis (temporal dynamics)
- Sequence length = 60 trading days
- PyTorch, 10,000 parameters

Features added

- EPS revision momentum
- Revision breadth
- Recommendation drift
- Earnings surprise history
- Consensus growth estimates

Result

- Fundamental fill rate: **0/13** tickers had $\geq 3/6$ fundamental features populated
- Mean model trust weight: **0.00**
- Forecast sources: 0 CNN-LSTM / 3 blend / 10 historical mean
- Raw CNN-LSTM outputs: **-4.5** to **+5.5** annualised (nonsensical)

What We Learned

Even with shrinkage and a more expressive architecture, the model could not clear the out-of-sample

Round 3 — Cross-Sectional Gradient Boosting

The Correct Framing

Instead of predicting each asset's *absolute* 21-day return (dominated by the unpredictable market factor), predict the *cross-sectional rank* of each asset's forward return within the universe. Strip out the market, keep only relative signal.

- Model: HistGradientBoostingRegressor
- Features: Van der Waerden rank-z scored, within each date
- Target: cross-sectional rank of 21-day forward log return
- Walk-forward split with 21-day embargo to prevent leakage
- Metric: Spearman Information Coefficient (IC), averaged per date
- Decision rule: trust model only if $t(\text{IC}) \geq 2.0$

Round 3 — The Apparent Breakthrough

Initial IC = +0.075, $t = 8.03$

Looks like a strong signal. Passes any reasonable threshold.

But it was wrong.

Where the numbers came from

466 daily IC observations. Standard t -statistic:

$$t = \frac{\overline{\text{IC}}}{\sigma_{\text{IC}}/\sqrt{n}} = \frac{0.075}{0.325/\sqrt{466}} \approx 8.03$$

Everything hinges on the assumption that $n = 466$ *independent* observations.

The Killer — Overlapping Windows

The Statistical Sin

Each date's IC uses a **21-day forward return**. Consecutive dates share 20 of 21 days of information. The 466 “observations” carry the information content of roughly $466/21 \approx 22$ truly independent observations.

Quantity	Naive	Overlap-corrected
Mean IC	+0.0087	+0.0087
Std IC	0.3248	0.3248
Effective n	466	22.2
t -statistic	+0.58	+0.13
Threshold for trust	2.0	2.0

Conclusion

The “signal” was entirely an artifact of an inflated sample size. With the correct effective n , the IC is statistically indistinguishable from zero. **The model has no skill.**

The modelling failure was compounded by engineering bugs that made intermediate results *look* plausible.

Four categories, each one masked by the synthetic-data fallback or by an unbounded output layer.

Bug #1 — Infinite Values in Feature Panel

Symptom

- `ValueError: Input X contains infinity`
- Raised inside `StandardScaler.transform`
- Crashed Stage 3 for every ticker

Root cause

- `vol_roc10 = volume.pct_change(10)`
- Bursa mid-caps have zero-volume days (trading halts)
- Division by zero $\rightarrow \pm\infty$

Why the synthetic path missed it

Synthetic volumes were generated as `np.abs(rng.normal(...))` — *strictly positive*. The bug could only surface on real data.

Fix

`feats.replace([np.inf, -np.inf], np.nan)`
at the end of `compute_indicators_for_ticker`.
Row-level NaN is handled by `dropna()`.

Bug #2 — Shariah Screening Vocabulary Mismatch

Symptom

- Maybank and Carlsberg *passed* the Shariah screen
- Both appear in the final portfolio
- A **compliance failure**, not just a bug

Root cause

- Live ingestion re-labels via Yahoo industry: "Excluded (brewers)"
- NON_COMPLIANT_SECTORS contains "Brewery"
- The two sets never intersect $\Rightarrow$ Tier 1 returns True unconditionally

Why it was hidden

The synthetic path copied sector labels verbatim from DEFAULT_UNIVERSE, so "Brewery" matched and Tier 1 *appeared* to work. The live path used a different vocabulary.

Fix

NON_COMPLIANT_INDUSTRY_KEYWORDS now maps Yahoo substrings *directly onto canonical labels*, with an import-time assertion that the two vocabularies stay in sync.

Bug #3 — Vol Z-Scored Before Being Used as Vol

Symptom

- Every ticker's `exp_vol_ann` printed as 0.0300 — the clip floor
- Optimiser piled into the highest-return tickers as if they were also the lowest-risk

Root cause

- `vol21_ann` was cross-sectionally z-scored
- Z-scored vol has mean ≈ 0 across tickers
- `clip(0, 0.03, 0.90) → 0.03` for every ticker

Fix

Capture raw per-ticker vol *before* the cross-sectional transform, in a dict keyed by ticker:

```
raw_vol_by_ticker =  
panel.groupby(...).mean().to_dict()
```

Meta-lesson

Cross-sectional z-scoring destroys the absolute scale of any feature. If a downstream stage needs the *original* value, snapshot it before the transform.

Bug #4 — Look-Ahead Bias in Fundamentals

The Problem

Fundamental features were fetched as a *single snapshot from today*, then broadcast across all five years of history:

$$\text{panel}[c] = \text{snap.get}(t, \text{NaN}) \quad \forall \text{dates}$$

So on 2021-01-04 the model saw the 2026 consensus EPS estimate.

- In a genuine backtest, this is leakage.
- It doesn't invalidate the negative result (the model failed anyway), but it *does* mean the IC evaluation is not a true OOS test.
- Any future attempt to build a working model must use a point-in-time database with proper as-of joins.

Resolution in Option A

The fundamental provider was removed from the default pipeline. It remains as a documented extension hook for anyone with a proper point-in-time source.

Three architectures tested. Two feature sets. Two different target formulations (absolute, cross-sectional).

All failed.

At some point, another architecture change is not a solution — it is a way of avoiding the finding.

We made a decision

Stop trying to forecast. Report trailing historical statistics instead, label them honestly as such, and let the optimiser work with what actually exists.

Option A — Honest Relabelling

What changed

- Stage 3 renamed: “Historical Return Estimation”
- Column names: `exp_return_ann` → `hist_return_ann`
- No model fitting in Stage 3
- Each ticker gets its own trailing mean return and realised vol

What stayed

- Full six-stage pipeline structure
- Shariah screening (Tier 1 + Tier 2)
- Mean-CVaR optimiser with transaction costs, holding caps, and zakat
- Investor profiling questionnaire
- Stress testing (reformulated — next slide)

Fixing the Stress Test

Before (broken)

- Multiply central return estimate by scenario fraction
- For a low-expected-return portfolio:

$$\begin{aligned}\text{GFC} &= 0.05 \times 3.84\% \\ &= \mathbf{+0.19\%}\end{aligned}$$

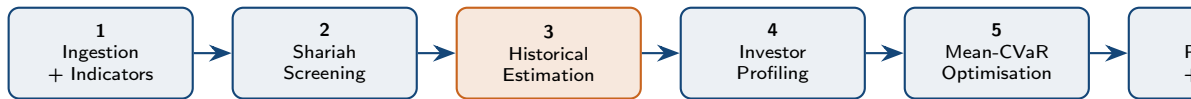
- Every scenario produces a *positive* return
- Not a stress test — a decoration

After (correct)

- Compute portfolio vol from covariance matrix:
 $\sigma_p = \sqrt{w^\top (\text{DCD}) w}$
- Apply σ -multiple drawdowns calibrated from historical peak-to-trough equity moves:

Scenario	σ multiple
2022 rate shock	1.25
COVID 2020	1.70
Dot-com 2000	2.45
GFC 2008	2.75

Final Architecture



- Stage 3 (highlighted) is now a *descriptive statistics* stage, not a forecasting stage
- Every output column is labelled with its source (`estimate_source = "historical_5y"`)
- No component in the pipeline claims predictive skill it does not have

Stage 3 — Historical Estimation Output

Ticker	hist_return_ann	hist_vol_ann	n_obs
5211.KL Sunway	+23.80%	24.64%	1,183
3336.KL IJM	+15.42%	29.91%	1,185
5285.KL SD Guthrie	+13.18%	25.12%	1,183
8869.KL Press Metal	+8.81%	27.68%	1,185
3816.KL MISC	+7.71%	16.55%	1,185
1961.KL IOI	+7.48%	20.27%	1,185
6033.KL Petronas Gas	+5.72%	14.79%	1,185
3026.KL Dutch Lady	+0.71%	15.21%	1,185
3689.KL F&N	+0.60%	21.37%	1,185
4707.KL Nestlé	−5.34%	17.75%	1,185
5183.KL Petronas Chemicals	−9.68%	33.27%	1,183
7113.KL Top Glove	−25.83%	48.94%	1,185
5168.KL Hartalega	−32.56%	48.88%	1,183

Return range [−32.56%, +23.80%]. Vol range [14.79%, 48.94%]. These are trailing five-year means, **not** forecasts.

Stage 5 — Optimal Portfolio

Ticker	Weight	Sector	hist ret.	hist vol.
5211.KL Sunway	30.00%	Conglomerates	23.80%	24.64%
3816.KL MISC	20.97%	Marine Shipping	7.71%	16.55%
5285.KL SD Guthrie	17.22%	Packaged Foods	13.18%	25.12%
6033.KL Petronas Gas	14.41%	Oil & Gas	5.72%	14.79%
3336.KL IJM	12.45%	Conglomerates	15.42%	29.91%
1961.KL IOI	4.96%	Packaged Foods	7.48%	20.27%

Portfolio-level metrics

Expected return (gross, annualised)	14.14%
Expected return (net of cost + zakat)	11.49%
CVaR @ 90% confidence	8.78%
Turnover from initial	100.00%
Transaction cost drag	0.15%
Annual zakat	2.50%

Stage 6 — Stress Test Results

Scenario	σ multiple	Portfolio return
2022 rate shock	1.25	-16.18%
COVID 2020	1.70	-22.01%
Dot-com 2000	2.45	-31.72%
GFC 2008	2.75	-35.60%

Why this is now credible

- Applied to the *portfolio's own* volatility ($\sigma_p = 12.95\%$)
- Scenario multiples are calibrated from real peak-to-trough equity drawdowns
- Signs are correct: every crisis produces a loss
- A client can see both the central estimate (+14.14%) and the crisis floor (−35.60%) side by side

Why AI Failed Here — Five Reasons

- ❶ **Sample size.** A 21-day horizon on 5 years of data gives ≈ 250 non-overlapping observations per ticker. A model with 10,000 parameters cannot be constrained by that.
- ❷ **Signal-to-noise ratio.** Real equity signals have $IC \approx 0.03\text{--}0.05$. Detecting that requires *hundreds* of independent cross-sections. Overlapping windows reduced our effective n by $\sqrt{21}$.
- ❸ **Feature quality.** Fundamental features (analyst estimates, revisions) were unavailable at retail data quality. **0/13** tickers had $\geq 3/6$ features populated.
- ❹ **The efficient-market null.** Twenty-one-day returns on liquid large-caps are essentially unpredictable from public technicals. This is not a bug — it is the correct finding.
- ❺ **Overfitting capacity.** MLP, CNN-LSTM, and gradient boosting all have more parameters than the data can constrain. All three collapsed to predicting the training mean.

When Is AI Deep Learning Viable?

Viable

- **Large N :** 10^4+ independent samples
- **Stationary signal:** language, vision, physical systems
- **Rich features:** high-dimensional, dense, well-populated
- **Low noise floor:** prediction error dominated by bias, not variance
- **Benchmark exists:** a naive baseline you can beat and measure

Not Viable

- **Small N :** a few dozen cross-sections
- **Efficient markets:** signal buried under noise
- **Sparse features:** retail-quality emerging-market data
- **High noise floor:** model fits the noise, not the signal
- **No benchmark:** can't tell a working model from an overfit one

Shariah-compliant equity forecasting on Bursa Malaysia checks **all five “not viable” boxes.**

*The most valuable thing a model can do
is tell you when not to use it.*

The shrinkage mechanism we built — blend model output with the historical mean, weighted by out-of-sample skill — **worked correctly**. It detected that all three architectures had *negative* skill and neutralised them.

The pipeline's job is not to be right. It is to be honest. On that metric it succeeded.

Engineering Lessons

1. Synthetic fallbacks hide live-data pathologies

Every live-only bug (infinite volume, zero-volume counters, missing fundamentals) was invisible on the synthetic path. If you have a fallback, you must test the live path *specifically*.

2. Vocabulary drift between layers is a compliance risk

The Shariah screening bug silently passed Maybank and Carlsberg into the final portfolio. Import-time assertions between the two vocabularies are now the only reason it cannot recur.

3. Cross-sectional transforms destroy absolute scale

Z-scoring a feature for the model also z-scores it for everything downstream. If a later stage needs the original value, snapshot it before the transform.

4. Overlapping windows inflate t -statistics by $\sqrt{\text{horizon}}$

Always use an effective- n correction or a Newey–West HAC estimator when evaluating forecasts on overlapping windows.

AI is not a universal solvent.

It requires:

- Data at a scale the problem does not naturally provide
- A signal-to-noise ratio the domain does not guarantee
- Features that only exist if you can afford them
- A benchmark that tells you when to stop

When those conditions fail,
**the correct engineering decision
is to stop, label honestly, and ship what works.**

Thank You

Questions?